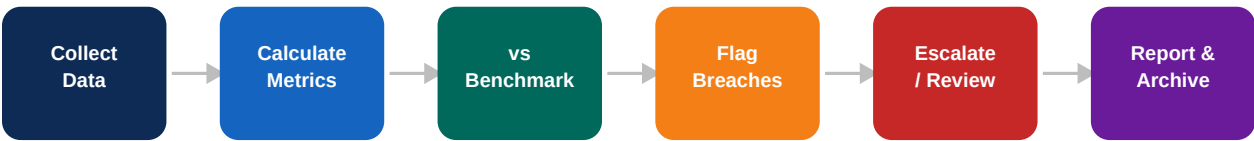


Portfolio Monitoring & Performance Evaluation Guide

Finance Risk & Investment Intelligence System Doc
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This guide defines performance metrics, monitoring procedures, benchmark comparison methods, volatility techniques, drawdown policies, and attribution frameworks for all portfolios in the Finance Risk & Investment Intelligence System.

Portfolio Monitoring Workflow



1. Core Portfolio Performance Metrics

All portfolios are evaluated against a comprehensive set of risk-adjusted and absolute return metrics. All metrics are calculated at the portfolio level and reported monthly to the Investment Committee:

Metric	Formula / Description	Low Target	Medium Target	High Target	Review
Total Return	NAV change + distributions	Inflation +1%	≥ Benchmark	Benchmark +2%	Daily
Sharpe Ratio	$(R_p - R_f) / \sigma_p$	≥ 0.6	≥ 0.8	≥ 1.0	Monthly
Sortino Ratio	$(R_p - R_f) / \sigma_{\text{downside}}$	≥ 0.8	≥ 1.0	≥ 1.2	Monthly
Alpha (annualised)	Portfolio return – benchmark	> +0.5%	> +1.0%	> +1.5%	Monthly
Beta	$\text{Cov}(R_p, R_m) / \text{Var}(R_m)$	0.4 – 0.7	0.7 – 1.0	0.9 – 1.3	Quarterly
Max Drawdown	Peak-to-trough portfolio loss	≤ 4%	≤ 8%	≤ 15%	Daily
Tracking Error	Std dev of active return	< 2%	< 5%	< 8%	Monthly
Information Ratio	Alpha / Tracking Error	≥ 0.3	≥ 0.4	≥ 0.5	Quarterly
Calmar Ratio	Annual return / Max drawdown	≥ 0.5	≥ 0.8	≥ 1.0	Quarterly

2. Benchmark Comparison Standards

Each portfolio must be assigned a primary benchmark reflecting its investment mandate. Benchmark selection, comparison methodology, and tracking error limits are governed by these standards:

Portfolio Type	Primary Benchmark	Secondary Benchmark	Tracking Error Limit	Rebalance Alignment
Global Equity (High)	MSCI World	MSCI ACWI	< 8%	Quarterly
Balanced Fund (Medium)	60/40 MSCI World / Barclays	Custom blended index	< 5%	Quarterly
Emerging Markets (High)	MSCI Emerging Markets	MSCI EM IMI	< 8%	Monthly
Bond Fund (Low)	Bloomberg Global Aggregate	ICE BofA IG Index	< 2%	Semi-annual
Money Market	3M T-Bill Rate	LIBOR / SOFR	< 0.5%	Daily

- **Relative Return:** Daily relative return ($R_p - R_b$) is calculated and tracked in a rolling 12-month chart for all portfolios.
- **Attribution Frequency:** Full attribution analysis (allocation, selection, interaction) must be completed monthly.
- **Custom Benchmarks:** Blended benchmarks for multi-asset funds require documented rationale and CIO written approval.
- **Annual Benchmark Review:** Benchmarks must be reviewed annually; changes require Investment Committee approval.

3. Volatility Measurement Techniques

Technique	Method	Parameters	Primary Use Case	Frequency
Historical Volatility	Std dev of log daily returns	Rolling 30/90-day window	Risk score input; VaR	Daily
EWMA	Exponentially weighted moving average	0.94 (RiskMetrics standard)	Short-term VaR calculation	Daily
GARCH(1,1)	Volatility clustering autoregressive	Rolling calibration window	1–5 day vol forecasting	Weekly
Implied Volatility	Derived from options market price	At-the-money options used	Market sentiment gauge	Daily
Realised Volatility	High-frequency intraday price data	5-min intervals, daily aggregated	Intraday risk monitoring	Daily
Conditional VaR (CVaR)	Expected loss beyond VaR threshold	95% and 99% confidence levels	Tail risk assessment	Monthly

4. Drawdown Analysis & Recovery Policy

Drawdown analysis measures peak-to-trough declines in portfolio value. When thresholds are breached, structured response procedures are triggered based on portfolio risk category:

Portfolio Type	Warning Threshold	Breach Threshold	Recovery Target	Breach Action	Review Authority
Low Risk	2.5%	4%	Within 90 days	Notify Risk Manager; review allocation	Portfolio Manager
Medium Risk	5%	8%	Within 120 days	Rebalance; risk review; commit review	Risk Manager + PM
High Risk	10%	15%	Within 180 days	Emergency committee; strategy review	Investment Committee

5. Performance Attribution Framework

Performance attribution decomposes portfolio returns into component sources using the Brinson-Hood-Beebower (BHB) model. Attribution reports are produced monthly:

Effect	Definition	Formula	Positive Signal	Reported
Allocation Effect	Return from over/underweighting sectors	$(R_{p_sector} - R_{b_total}) \times (w_{sector} - w_{benchmark})$	Overweight winning sectors	Monthly
Selection Effect	Return from picking securities within sectors	$(R_{p_security} - R_{b_sector}) \times w_{sector}$	Outperforming benchmark within sector	Monthly
Interaction Effect	Combined impact of allocation & selection	$(R_{p_security} - R_{b_sector}) \times (w_{sector} - w_{benchmark})$	Both active bets align positively	Quarterly
Currency Effect	Return from FX movements on foreign assets	Unhedged return difference	Favourable currency movements	Monthly
Timing Effect	Return from tactical rebalancing	Active timing vs systematic rebalance return	Timely risk reduction or capture	Quarterly

6. Performance Monitoring Alerts & Escalation

Alert Type	Trigger Condition	Automatic Action	Human Review Required
Sharpe Ratio Alert	Sharpe < target for 2 consecutive months	Flag on risk dashboard	PM submits PIP within 10 days
Drawdown Warning	Portfolio declines past warning threshold	Email to PM and Risk Manager	Review within 48 hours
Drawdown Breach	Portfolio exceeds hard breach threshold	Emergency alert to committee	Meeting within 24 hours
Alpha Decay Alert	Rolling 3M alpha < 0% vs benchmark	Watchlist flag generated	Attribution analysis required
Benchmark Deviation	Tracking error exceeds defined limit	System warning message generated	Rebalance review initiated
Volatility Spike	30-day vol 50% above historical average	Risk score auto-adjusted upward	PM notified within 2 hours

■ Synthetic data for educational simulation only. Not real investment advice.

7. Key Performance Indicator Dashboard — Reference Template

The FRIIS performance dashboard displays the following KPIs in real time. All metrics map directly to the SQL tables and can be queried by the AI intelligence agents:

KPI	SQL Source	Target Range	Alert Threshold	Update Freq.
Portfolio Total Return	portfolio_holdings + market data	Per mandate target	Below benchmark 3M	Daily
Aggregate Risk Score	asset_risk_logs.risk_score (weighted avg)	55 for medium portfolio	Score > 70	Daily
Escalation Flag Count	COUNT(escalation_flag=TRUE)	Zero open flags	Any open flag	Real-time
Sector Concentration	SUM(allocation%) GROUP BY sector	< 25% per sector	Breach of sector limit	Daily
Approved Decisions %	investment_decisions.approval_status	100% approved	Any 'Under Review' > 5 days	Daily
Drawdown Level	Derived from market_value changes	Within mandate limit	Warning threshold hit	Daily
Market Indicator Trend	market_indicators.trend_direction	Stable or improving	2+ rising risk indicators	Weekly

8. Monitoring Roles, Responsibilities & Escalation Contacts

The following table defines monitoring ownership and escalation contacts for each performance and risk metric category:

Metric Category	Primary Monitor	Alert Recipient	Escalation Contact	Review Cycle
Return vs Benchmark	Portfolio Manager	Investment Committee	CIO	Monthly
Risk Score Monitoring	Risk Manager	Portfolio Manager	CRO	Daily
Drawdown Tracking	Risk Manager + PM	Investment Committee	CIO + Board	Daily
Regulatory Compliance	Compliance Officer	Risk Manager	Legal + Board	As triggered
Liquidity Position	Treasury / Finance	CRO	Board Finance Cmte	Weekly
ESG Score Monitoring	ESG Analyst	PM + Compliance	Board Risk Committee	Quarterly